

ARUN ARUNACHALAM

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PROFESSIONAL SUMMARY

Identity-aware platform architect with 17+ years designing and modernizing mission-critical, cloud-native systems at enterprise scale within capital markets and financial services. Deep expertise in distributed systems, event-driven architecture, and AI-enabled platforms — with a builder's mindset and a track record of executing with speed and urgency on high-stakes, real-world infrastructure. Combines hands-on engineering depth in Java, Python, Kafka, Kubernetes, and AWS with applied GenAI (RAG pipelines, agentic AI) to unlock intelligent, secure, and scalable platform outcomes. Proven ability to own complex architectural decisions end-to-end and deliver career-defining platform transformations that move at the pace of the mission.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

Cloud-Native Architecture (AWS, Kubernetes) | Distributed Systems & High-Availability Platforms | Event-Driven Systems (Kafka) | AI & GenAI Platforms (RAG, Agentic AI) | Platform Modernization & Legacy Transformation | Data Engineering (Spark, Databricks, ETL/ELT) | Trading & Risk Platforms | Regulatory & Compliance Systems (FRB, FINRA, SEC) | Executive Stakeholder Leadership | Global Engineering Leadership

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Architected cloud-native, event-driven platform transformation (AWS, Kafka, Kubernetes) supporting **\$1T+ daily processing** — owning end-to-end architectural decisions across distributed, high-availability systems
- Modernized legacy infrastructure into scalable, resilient microservices-based platforms, improving throughput by **40%** and achieving **99.97% uptime** across 50+ systems
- Delivered real-time trading, pricing, and risk platforms with strict latency, reliability, and compliance requirements — operating under FRB, FINRA, and SEC regulatory mandates
- Implemented AI-assisted development workflows and ML-based anomaly detection, accelerating delivery velocity by **15%**
- Drove platform security and compliance posture across mission-critical systems, ensuring auditability and regulatory alignment at enterprise scale
- Built and led global engineering teams (**30+ across US/UK/India**) executing complex, multi-year platform programs with speed and urgency

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Architected and delivered real-time pricing and analytics platforms for structured products and mortgage trading, requiring high-throughput, low-latency distributed system design
- Led consolidation of risk systems contributing to ~\$2B capital efficiency, owning architectural decisions across integrated data and computation pipelines

Bank of America — AVP, Whole Loan Trading Desk Support (*prior to 2011*)

- Architected and delivered end-to-end Java EE trading support applications, establishing foundational platform patterns for reliability and scalability
- Introduced CI/CD practices reducing release cycle from monthly to weekly — embedding engineering excellence and delivery speed into the SDLC

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building GenAI platforms (RAG pipelines, agentic AI systems) for real-world, production-grade applications — applying AI to unlock intelligent decision systems at scale
- Applied ML for anomaly detection and operational intelligence across high-stakes financial infrastructure
- Driving AI-assisted SDLC workflows improving delivery speed, quality, and engineering productivity

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Architecture: Cloud-native architecture (AWS, Kubernetes), distributed systems, event-driven systems (Kafka), enterprise architecture, high-availability platforms (99.97% uptime), microservices, REST/SOA

AI & Data: GenAI (RAG pipelines, agentic AI), ML anomaly detection, AI-assisted SDLC, Spark, Databricks, ETL/ELT, Snowflake, Hadoop, Cassandra, MongoDB

Engineering: Java, Python, Node.js, SQL, Oracle, Sybase, WebLogic, Kafka, Kubernetes, AWS, Git

Compliance & Governance: FRB, FINRA, SEC regulatory compliance, SOX, ITIL, Agile, SDLC

Leadership & Delivery: Executive stakeholder management, global team leadership (30+ engineers), P&L ownership, strategic planning, program management, ServiceNow, JIRA, Visio, Microsoft Project